

Device-Independent Quantum Communication Technology

From Theory to Hardware Security

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Agenda

- **The Security Gap:** Quantum Promise vs. Physical Reality
- **Theoretical Foundations:** EPR Paradox and Bell's Theorem
- **Hardware Architecture:** Engineering Entanglement
- **Vulnerabilities:** The APD Blinding Exploit
- **The Solution:** Device-Independent QKD (DI-QKD)
- **True Randomness:** QRNGs and Bias Extraction

The Quantum Promise vs. Physical Reality

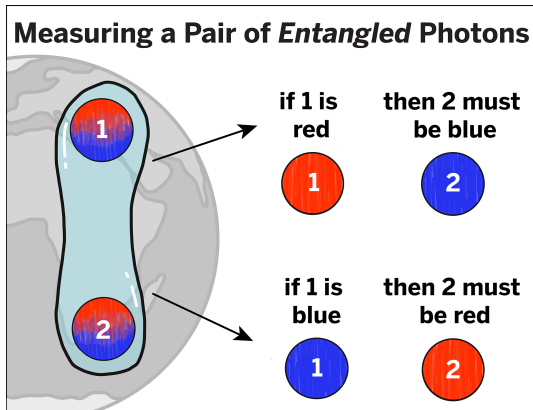
- **Information-Theoretic Security:** Guaranteed by No-Cloning and Wave-function Collapse.
- **The Physical Gap:** Ideal proofs ignore fiber loss ($\alpha \approx 0.2$ dB/km) and hardware imperfections.
- **Security Collapse:** If detection hardware is compromised, the mathematical model of the quantum state becomes invalid.

Shor-Preskill Threshold

Secure key extraction is only possible if the Quantum Bit Error Rate (QBER) remains below $\approx 11\%$.

Quantum Entanglement

- **Definition:** A phenomenon where particles become intrinsically linked, such that the quantum state of one is inseparable from the others.
- **Non-Local Correlation:** Measuring one particle instantaneously determines the state of its entangled partner, regardless of physical distance.
- **Superposition Collapse:** The act of measurement collapses the shared wave-function simultaneously.



EPR Paradox and the CHSH Inequality

- **Local Realism:** Einstein's argument for "Hidden Variables" to avoid "spooky action".
- **CHSH Inequality (1964):** Tests correlations between Alice and Bob at varying angles.
- **Realism vs. QM:** Violation proves measurement actively defines reality.

Mathematical Bounds

Classical Hidden Variables: $S \leq 2$

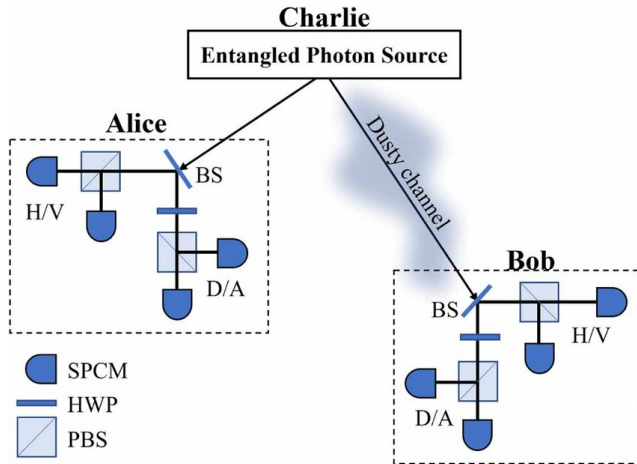
Tsirelson's Bound (Quantum): $S = 2\sqrt{2} \approx 2.828$

Bell State:

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|HH\rangle + |VV\rangle)$$

Higher S values indicate stronger non-classical correlation.

Engineering Entanglement: The BBM92 Setup



The 4-Step Engineering Process

- 1 Generation:** SPDC via non-linear crystals (BBO/ppKTP).
- 2 Superposition:** Path overlap to erase "which-path" info.
- 3 Compensation:** Fixing phase errors via YVO_4 .
- 4 Splitting:** Wedge mirrors separate signal/idler.

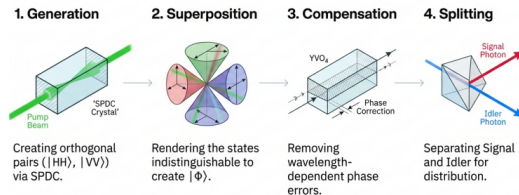
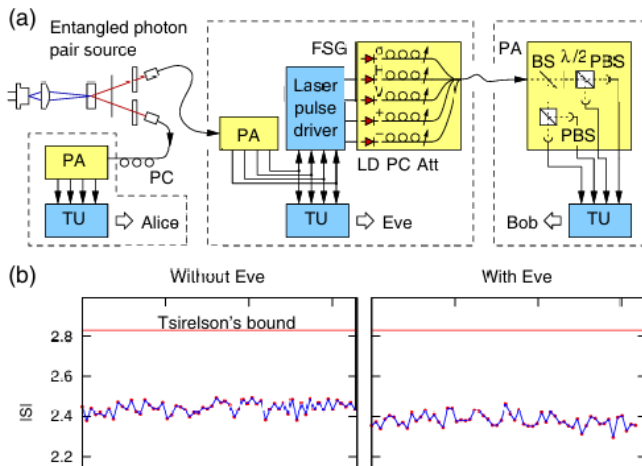


Figure: Entanglement Generation Process.

The Detector Blinding Exploit (Gerhardt et al.)



Mechanism of the Faked-State Attack

Phase 1: Blinding

- Eve injects a continuous, bright, circularly polarized classical laser into the receiver.
- This overloads Avalanche Photodiodes (APDs), dropping them below breakdown voltage.
- APDs exit Geiger mode, becoming entirely **blind to single photons**.

Phase 2: Faking the Violation

- Eve intercepts and measures the genuine quantum state.
- She fires a tailored, **high-intensity classical pulse** at the blinded detectors.
- This macroscopic pulse forces the targeted APD's current over threshold, artificially triggering a "click".

Countermeasures: Defending the Detectors

- **High-Energy Laser Detection:** Integrating an optical power meter or dedicated watchdog detector at the receiver's entrance.
- **Continuous Monitoring:** The watchdog continuously profiles incident optical power against the expected single-photon threshold.
- **Abort Protocol:** If power exceeds the threshold, the system flags a blinding laser attack and immediately aborts key distribution.
- **The "Arms Race":** Physical patches often lead to new, creative quantum hacking techniques, underscoring the need for hardware-independent security (DI-QKD).

Device-Independent QKD: The Zero-Trust Model

- **Black Box Approach:** Treat all hardware as untrusted; security relies on statistical outputs.
- **Verification:** Use the Bell test directly on hardware. $S > 2$ ensures security.
- **Monogamy of Entanglement:** A third party (Eve) cannot correlate with a maximally entangled pair.

Paradigm Shift

Moves from modeling the **internal physics** of a device to verifying the **external correlations**.

Closing the Loophole: QRNGs

- **Unpredictability:** DI-QKD fails if Eve predicts measurement bases.
- **Quantum Randomness:** Spatial superposition in a 50/50 beam splitter.
- **Von Neumann Extractor:** Eliminates hardware bias by processing non-overlapping bit pairs.

Extractor Logic

Input: '01' → Output: 0

Input: '10' → Output: 1

Discard: '00' and '11'

Conclusion

- Physical layer vulnerabilities (APDs) compromise standard QKD.
- DI-QKD provides security even when devices are fundamentally compromised.
- Future networks must integrate robust Bell tests and bias-extracted QRNGs.

Thank You!

Questions?

Simulation: <https://princekumariofficial.github.io/rng-von-neumann-extractor/>

References



Quantum Atlas, University of Maryland.
Entanglement.



ResearchGate.
BBM92 protocol block diagram.